NNN-B-1505A
September 10, 1976
SUPERSEDING
Int. Fed. Spec. NNN-B-001505(GSA-FSS)
September 17, 1969 and
Fed. Spec. DD-B-610
February 8, 1955

FEDERAL SPECIFICATION

BOTTLES, SCREWCAP, GLASS

(LABORATORY USE)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. These glass bottles are intended for miscellaneous uses, such as liquid and tablet containers and sample vials.

1.2 Classification.

1.2.1 Types and styles. Screw-neck bottles with screwcaps, covered by this specification shall be of the following types and styles as specified:

Type I. Bottles, screw-neck, narrow mouth.

Style I-Round.

4 milliliters (ml.)
8 ml.
9 ml.
12 ml.
16 ml.
20 ml.
23 ml.
60 ml. (2fl. oz.)
120 ml.

FSC 6640

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Style II-Square.

250 ml.

Type II. Bottles, screw-neck, wide mouth.

Style I-Round.

20 ml. (5 drams)
84 ml.
130 ml.
210 ml.
236 ml. (8 fl. oz.)

1.2.2 Grades of glass. Screwcap bottles shall be furnished in the following grades of glass as specified:

Grade A-Borosilicate glass.
Grade B-Soda-lime glass.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

Federal Specifications:

PPP-B-566 -- Boxes, Folding, Paperboard.
PPP-B-636 -- Boxes, Shipping, Fiberboard.
PPP-B-676 -- Boxes, Setup.

Federal Standards:

Fed. Std. No. 66 -- Steel: Chemical Composition and Hardenability.

Fed. Std. No. 123 -- Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issue, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

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(Federal Government activities may obtain copies of Federal Specifications and Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 -- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 -- Marking for Shipment and Storage.

(Copies of Military Specifications, Standards and Drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

American Society for Testing and Materials (ASTM) Standards:

C-148 -- Polariscopic Examination of Glass Containers.
C-149 -- Thermal Shock Test on Glass Containers.
C-162 -- Glass and Glass Products, Definition of Terms Relating to.
E-438 -- Glass in Laboratory Apparatus, Specification for.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Glass.

3.1.1.1 Grade A. Grade A (borosilicate) glass shall conform to the requirements for type 1 of ASTM E-438.

3.1.1.2 Grade B. Grade B (soda lime) glass shall conform to the requirements for type II of ASTM E-438.

3.1.2 Plastic. The material for the screwcaps shall be heat-resistant phenol-formaldehyde plastic that is free from cellulose acetate and urea compound. As an alternate, material for the type II bottles may be in accordance with 3.2.2.2.

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3.2 Design and construction. The bottles shall be made of transparent and colorless glass. The bottles shall comply with the requirements of 3.2.1 and, unless otherwise specified (see 6.1), shall be supplied with the bottles and closure caps assembled. The closure caps shall comply with the requirements of 3.2.2.

3.2.1 Bottles.

3.2.1.1 Shape.

3.2.1.1.1 Types I and II, style I bottles. The style I bottles (types I and II) shall be conventional cylindrical bottles with exterior threaded finishes to accommodate a screwcap closure. A straight section on the top of the bottle shall be permitted between the take-out bead and the start of the shoulder.

3.2.1.1.2 Type I, style II bottles. The style II bottles (type I) shall have a square base, parallel rectangular faces and an exterior threaded finish to accommodate a screwcap closure. A straight section on the top of the bottle shall be permitted between the take-out bead and the start of the shoulder.

3.2.1.2 Dimensions. The dimensions and finish of the bottles shall conform to table I.

TABLE I. Dimensions of screwcap bottles

Type I. Bottles, screw-neck, narrow mouth						
Capacity ml.	Height of Bottle mm. ± 1.6 mm.	Diameter of Bottle mm. ± 1.6 mm.	Mouth I.D. minimum mm.	G.C.M.I. 1/ Thread E 2/	G.C.M.I. 1/ Screw Finish	Cap No.
4 (1 dram)	45	15	8	13	425	13
8 (2 dram)	60	17	8.7	15	425	15
9	60	20	8.7	15	425	15
12 (3 dram)	65	19	8.7	15	425	15
16 (4 dram)	70	21	11.0	18	400	18
20 (5 dram)	75	20	11.0	18	400	18
23	80	25	12.4	22	400	22
60 (2 fl. oz.)	83.4	40.4	12.4	22	400	22
120 (4 fl. oz.)	115	49	12.4	22	400	22
250	159	50 x 50* *square	15.6	28	400	28

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TABLE I. Dimensions of Screwcap bottles. (Con.)

TYPE II. Bottles, screw-neck wide mouth

Capacity ml.	Height of Bottle mm. ± 1.6 mm.	Diameter of Bottle mm. ± 1.6 mm	Mouth I.D. minimum mm.	G.C.M.I. 1/		Cap No.
				Thread E 2/	Screw Finish	
20 (5 drams)	61	27.5	12.4	22	400	22
84	70	55	35.1	48	400	48
130	102	51	35.1	48	400	48
210	143	55	35.1	48	400	48
236 (8 fl. oz.)	129* *+1.6,-2.0	59.5* *+2.0,-1.6	44.0	58	400	58

1/ G.C.M.I. is for Glass Container Manufacturers' Institute.
2/ Screw thread is the screw diameter in millimeters.

3.2.1.3 Identification marking. At the option of the manufacturer, the name or trademark of the manufacturer may be permanently marked on each bottle.

3.2.2 Screwcaps.

3.2.2.1 Plastic material. Screwcaps for types I and II bottles shall be fabricated from autoclavable and impact-resistant plastic as indicated in 3.1.2 or 3.2.2.2.

3.2.2.2 Liners. For type I bottles, a liner shall be secured to the inside of each cap in such a manner that the liner does not fall out of the cap or is not off-center when a screwcap is secured to a bottle. Each liner shall be supplied with a facing of sufficient thickness and resilience to meet the requirements of 3.3. The adhesives and materials used for the liner and facing shall be autoclavable (see 3.3.2). For type II bottles, liners shall be as specified above, or shall be furnished as linerless, one-piece polypropylene screwcaps that meet the requirements of 3.3.

3.3 Assembled bottles.

3.3.1 Leakage. The bottles shall show no evidence of leakage when tested as specified in 4.4.1.

3.3.2 Autoclavability. When tested as specified in 4.4.2, the bottles and screwcaps shall show no evidence of the following:

a. Bottles breaking, cracking or chipping.
b. Screwcaps visibly turning color, deteriorating or deforming.
c. Screwcap liners supplied are detached (or detachable) from cap or otherwise unsuitable for use.

3.4 Physical characteristics (bottles).

3.4.1 Strain. Bottles shall be so annealed that, when tested as specified in 4.4.3, they shall show no strain greater than temper no. 4.

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3.4.2 Thermal shock. When tested as specified in 4.4.4, the bottles shall not break, crack, or chip.

3.5 Workmanship. Workmanship shall be first class throughout. The bottles and screwcaps shall be free from defects which may impair their serviceability. The bottles and screwcaps shall be supplied clean, dry and ready for use. The bottles and screwcaps shall be free of deformation, cracks, chips, sharp surfaces, or sharp edges. The bottles shall be free of rocking and wobbling when placed on a surface that is smooth, flat, and level. The bottles shall be essentially uniform in thickness and shall be essentially free from an optical wave effect due to uneven glass distribution. The bottles may contain scattered seeds provided the number does not exceed five.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificate of quality. Certificate of quality, supplied by the manufacturer of the bottle, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the item identification, the name of the component, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

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4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and table II. The unit of product for inspection purposes shall be the individual bottle and screwcap.

TABLE II. Sampling for examination

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	1.0
Minor defects	II	2.5
For dimensional examination	S-2	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and table III. The unit of product for testing purposes shall be the individual bottle and screwcap (where applicable). Where possible, one sample may be used for testing two or more characteristics.

TABLE III. Sampling for tests

Characteristic	Requirement	Test procedure	Sample unit	Inspection level	AQL
Leakage	3.3.1	4.4.1	1 bottle w/cap	S-1	*
Autoclavability	3.3.2	4.4.2	1 bottle w/cap	S-1	*
Strain	3.4.1	4.4.3	1 bottle	S-2	*
Thermal shock	3.4.2	4.4.4	1 bottle	S-1	*

*Acceptance number shall be zero for all sample sizes.

4.3 Examination. The bottles shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with table IV. Examination shall not be restricted to the possible defects listed in the table.

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TABLE IV. Classification of defects

Categories	Defects *
Major	
101	Bottle not free of cracks, chips and breaks.
102	Bottle not free of blisters over 3/32 in.
103	Bottle not free of stones over 1/32 in.
104	Bottle not stable when placed on a flat and level surface.
105	Bottle not free of checks, crizzles and tears.
106	Bottle not free of open blisters and blisters over 1/32 in.
107	Bottle not free of localized thin areas.
108	Bottle not free of sharp edges.
109	Plastic cap not free of cracks, chips and breaks.
110	Bottle and screwcap not assembled when specified (see 3.2).
111	Screwcap liner missing and not securely attached.
Minor	
201	Bottle not clean and dry.
202	Bottle not free of blisters 1/16 in. to 3/32 in.
203	Surfaces not free of protrusions.
204	Bottle not free of rough seams and edges.
205	Bottle not free of deformed areas.
206	Bottle not smoothly finished.
207	Bottle not free of seeds in excess of five in number.

* Definitions of glass defects shall be in accordance with ASTM C-162.

4.3.2 Dimensional examination. The bottles shall be examined for defects in dimensions. Any dimension not within specified limits shall be classified as a defect.

4.3.3 Examination of preparation for delivery. An examination shall be made to determine compliance with the packaging, packing and marking requirements of section 5. Defects shall be scored as specified in table V. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 with AQL of 4.0 expressed in terms of percent defective.

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TABLE V. Examination of preparation for delivery

Examine	Defect
Contents	Not as specified.
Container	Not as specified.
Marking	Omitted; incorrect; illegible; improper size, location, sequence or method of application.
Material	Component missing or damaged.
Workmanship	Bulging or distortion of containers.

4.3.3.1 Examination of closure, waterproofing and banding of container. When the shipping container is required to comply with PPP-B-636, the examination for defects in closure, waterproofing or banding shall be in accordance with the appendix of the box specification.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Leakage. The bottle shall be filled with water, and the screwcap shall be securely applied handtight to at least the manufacturer's recommended application torque. The filled bottle shall be suspended in an inverted position for 30 minutes and checked for any evidence of leakage.

4.4.2 Autoclavability. The bottle with screwcap applied shall be subjected to three consecutive 30-minute autoclaving cycles at 250° Fahrenheit (15 psi). Between each autoclaving cycle, the capped bottle shall be removed from the autoclaving chamber and allowed to cool for 30 minutes.

4.4.3 Strain. The bottle shall be tested for strain in accordance with ASTM C-148, method A or B.

4.4.4 Thermal shock. The bottle shall be tested for thermal shock in accordance with either of the following methods:

a. Method A. The bottle shall be tested in accordance with ASTM C-149. The temperature differential between the hot and cold bath shall be 108°F. (42.2° Centigrade), and the maximum transfer time shall be 15 seconds.

b. Method B. The bottle shall be placed in an oven having an operating temperature of 60° ± 1°C. (140° ± 2°F). After 1 hour the bottle shall be transferred to a bath of ice water that is maintained at a temperature of 0°C. (32°F).

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5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B or C as specified (see 6.1).

TABLE VI. Packaging and packing quantities

Type	Style	Size (ml)	Number of bottles in each unit package	Number of bottles in each intermediate package	The number units of issue to be contained in each exterior container
I	I	4	48	None	18
I	I	8	48	None	18
I	I	9	1 (optional)	48	864
I	I	12	48	None	18
I	I	16	48	None	18
I	I	20	72	None	6
I	I	23	72	None	6
I	I	60	12	None	12
I	I	120	12	None	12
I	II	250	12	None	6
II	I	84	12	None	12
II	I	130	12	None	12
II	I	210	12	None	6
II	I	20	12	None	12
II	I	236	12	None	6

5.1.1 Level A or B.

5.1.1.1 Unit package.

5.1.1.1.1 Type I, 9 ml. At the option of the contractor, a unit package may be furnished for each 9 ml. bottle. If a unit package is furnished, each bottle shall be packaged in a box of an appropriate size constructed in accordance with PPP-B-566 or PPP-B-676. Closure of the box shall be adequate to prevent accidental opening.

5.1.1.1.2 Types I, and II, 4 ml. through 250 ml. capacity, except type I, 9 ml. The number of bottles specified in table VI, of one type, style, and capacity shall be packaged in a snug-fitting box constructed in accordance with PPP-B-636, type CF, class domestic, having a minimum bursting strength of 200 pounds. The box shall contain full-height half-slotted style partitions fabricated of chipboard or of the same material as the box. The partitions shall form an individual snug-fitting cell for each bottle. When

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bottles are stacked in two or more tiers, a fiberboard pad shall be interposed between each tier. Closure of the box shall be adequate to prevent accidental opening.

5.1.1.1.3 Intermediate package. The number of bottles specified in table VI of one type, style, and capacity shall be packaged as specified in 5.1.1.1.2, except that partitions shall not be required when unit packages are furnished.

5.1.2 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.1).

5.2.1 Level A. The number of unit packages specified in table VI of one type, style and capacity shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-636, type CF, class weather-resistant. Closure, strapping and waterproofing shall be as specified in the appendix of the box specification.

5.2.2 Level B. The number of unit packages specified in table VI of one type, style and capacity only shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-636, type CF, class domestic. Closure shall be as specified in the appendix of the box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Civil agencies. Marking for shipments shall be in accordance with Fed. Std. No. 123.

5.3.2 Military activities.

5.3.2.1 Interior packages. Each unit and intermediate package shall be marked in accordance with MIL-STD-129. Contract or purchase order number and name of contractor shall be shown.

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5.3.2.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129. In addition, the following legend shall be shown:

"GLASS - HANDLE WITH CARE."

6. NOTES

6.1 Ordering data. Purchaser should select the preferred options permitted herein, and include the following information in procurement documents:

a. Title, number and date of this specification.
b. Type, style and capacity required (see 1.2.1).
c. Grade of glass required (see 1.2.2).
d. Quantity required.
e. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
f. When closure caps are not required to be assembled to bottle (see 3.2).

6.2 This specification does not cover all types, classes, grades or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types, classes, grades or sizes normally procured to meet Federal requirements.

Military Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
DSA-DM

Project No. 6640-0949

Civil Agency Coordinating Activities:
GSA-FSS
HEW-NIH
USDA-AFS
VA-DMS

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DOCUMENT IDENTIFIER AND TITLE

NNN-B-1505A - BOTTLES, SCREWCAP (LABORATORY USE)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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DATE

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